Numerical solution of a PDE arising from prediction with expert advice

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Dedication

To my grandparents, with love

Abstract

This work investigates the online machine learning problem of prediction with expert advice through numerical analysis of a related PDE. The problem is a repeated two-person game involving decision-making at each step informed by n experts with geometric stopping condition; the continuum limit of the consequences of this game over a large number of steps leads to an elliptic PDE. This work presents a numerical scheme that allows us to solve this PDE for general number of experts n, and gives numerical results for $n \leq 8$.

Contents

Acknowledgements **i**

Dedication **ii**

Abstract **iii**

List of Tables **vi**

List of Figures **vii**

1 Introduction **1**

2 Background **5**

3 Analysis of a General Numerical Scheme **8**

3.1 Analysis of a general numerical scheme 8

3.1.1 Existence and uniqueness 9

3.1.2 Properties of solutions 13

3.1.3 Convergence rates 18

3.1.4 Restricting the domain 21

4 Implementation of the Numerical Scheme **24**

4.1 Implementation of the Numerical Scheme 24

4.1.1 Numerical Scheme 24

4.1.2 Domain Reduction 26

4.1.3 Flattening the Domain 30

4.1.4 Boundary Conditions 31

4.1.5 Techniques for improving efficiency 32

5 Numerical Experiments **34**
5.1 Numerical experiments 34
5.1.1 Evidence for Convergence of the Scheme 34
5.1.2 Optimal Strategies 35

6 Conclusion **43**

References **44**

Appendix A. Glossary and Acronyms **48**
A.1 Definition of viscosity solution 48

List of Tables

4.1 Total number of gridpoints in the original domain vs. the reduced domain. . . 30
5.1 Slopes for 3 experts. This supports the earlier finding that strategies 2 and 3 are optimal. 39
5.2 Slopes for 4 experts. This supports the earlier finding that strategies 5 and 6 are optimal,and presents the possibility that strategies 4 and 7 may be optimal, or may be converging to slightly-suboptimal levels. 40
5.3 Slopes for 5 experts. This indicates that several strategies from 8 to 15 may either be globally optimal, or converging to a level close to optimality. 40
5.4 Slopes for 6 experts. This indicates potential optimality of strategies 20, 22-24, 26, 27. 41
5.5 Slopes for 7 experts. This indicates potential optimality of ths trategies 40, 42, 44, 46, 48, 50, 52, 54. 42

List of Figures

5.1 In both of these plots, we can see that maximum error scales linearly with h^2. 35

5.2 In both of these plots, we can see that the boundary condition affects the solution, but only within a percentage of gridpoints close to the boundary. . . 36

5.3 37

5.4 37

5.5 38

5.6 38

5.7 39

Chapter 1

Introduction

Structural challenges in handling data directly influence the methods of machine learning that can be used to analyze a data set sensibly. Methods in batch learning assume that the test data will be similar to training data: in other words, they assume that the data takes on a fairly static nature. In these situations, one usually assumes that the training and test data are independent and identically distributed based on some probability distribution, so a model is evaluated on how well it can learn this distribution and make relevant predictions based on it in order to minimize some expected loss. However, many settings require a different approach, since the assumptions of batch learning no longer apply - e.g., settings in which data is constantly changing, and/or in which it is not possible to store all the data in RAM at once. Online learning methods, such as the ones in this work, seek to make predictions in scenarios where the landscape is constantly changing, even in the extreme case in which there are no correlations at all between data collected at different time steps. In this case, there is no stationary probability distribution present, so the goal shifts from minimizing expected loss (doing well on an absolute scale) to minimizing regret (doing well relative to the best agent in the system).

A classical question in online learning is how to make predictions with expert advice. Given the advice of n experts who each make a prediction at each time step t, a player observes the experts' predictions and decides whose advice to follow. This problem originated in the works of Cover [1] and Hannan [2]. At a given time step, the experts may make predictions of 1 or 0 signaling their guess that it is or isn't, respectively, a good idea to invest in a commodity at that time. Based on the predictions, the player then decides whether or not to invest, then the experts and player incur individual losses depending on whose guesses were correct. The outcomes of the market are based on a probability distribution

p which is not generally stationary with respect to time. In the absence of a stationary probability distribution, the player's goal shifts from minimizing their expected loss (doing well on an absolute scale) to minimizing their regret (doing well compared to the other agents). One common choice in the literature is linear regret, which measures how badly the player did, relative to the agent with the overall best choices. Goals within this problem are to find the optimal algorithm, adversary (also referred to as "market"), and regret in both settings of this problem: the "geometric horizon" (GH) setting, in which the process has a constant probability of halting at each time step, and the "finite horizon" (FH) setting, in which the game is played for a predetermined number T of time steps. This work focuses on the geometric horizon (GH) setting of the problem. Applications of prediction from expert advice include algorithm boosting [3], stock price prediction and portfolio optimization [3], self-driving car software [4], and many other problems.

The multiplicative weights algorithm for the FH problem [**?**] provides a bound on the regret which is tight in the limit of large n and T, but is sub-optimal for the numbers of experts that appear in practice. For $n \leq 3$, optimal and quasi-optimal algorithms for the GH and FH problems, respectively, have been developed by Gravin et al. [5] and Abbasi et al. [6]. Additionally, lower bounds for regrets in a broader class than multiplicative weights has been shown [7]. However, extending these methods to more generality is said to be combinatorially difficult, motivating a different approach. Additionally, a PDE approach similar to ours has been taken and a solution has been developed for the four-expert GH problem [8]; however, the methods relied on guessing and verifying a solution, so are not directly transferable to larger numbers of experts. Additionally, it has been proven that a strategy known as COMB is optimal in the 4-expert case [8] and experimental evidence indicates that it is not optimal in the 5-expert GH case [9]. Further work on algorithms that work well in practice has been studied both for the finite and geometric horizon problems [6,10–14]. In summary, the theory is mostly limited to the case of five or fewer experts: beyond this point, nothing is known about optimal strategies or regret.

Drenska [15–17] views the steps of prediction with expert advice as the discrete analogue to a value function, and uses this to form a PDE. Repeated two-player games also appear in the PDE literature in multiple other settings [18–29]. In order to analyze the long-term behavior of the game, the value function is scaled, where the scaled value functions converge to the viscosity solution of a PDE. In the limit of large t, the scaled value functions converge to the viscosity solution of a PDE, which provides a description of the game if it is played for sufficiently long. The solution of the discrete problem in the GH case for n experts converges

to a viscosity solution of the elliptic equation

$$u - \frac{1}{2}\max_{\mathbf{v}\in\{0,1\}^n} \mathbf{v}^T \nabla^2 u \mathbf{v} = g \qquad \text{on } \mathbb{R}^n \tag{1.0.1}$$

where u is the continuous analog of the value function and g is the payoff function at the end of the game. In the cases $n = 2$ and $n = 3$, closed-form solutions of the elliptic PDE were found [15], agreeing with previous results [5]. The solution was later found for $n = 4$ experts as well [8]. An analogous parabolic PDE exists for the finite-horizon setting of this problem [15], motivating a similar treatment in terms of numerics for the parabolic equation as well: progress for the FH/parabolic case can be found in [30]. Further, related settings such as prediction against a limited adversary [31] (rather than the optimal adversary being studied in this work) and malicious experts [32] have been studied as well. A related PDE where v is chosen just from the standard basis vectors $e_1, ..., e_n$ has been studied and has a solution for a general number n of experts [33, 34]. Additionally, some PDE approaches to online learning problems and neural network-based problems can be found in [35]. However, the ability to solve these cases explicitly relied on symmetries in the problem that do not obviously exist for higher n in the standard elliptic equation, motivating a numerical approach.

The known solutions of the PDE 1.0.1 are presented here for the case

$$g(x_1, ..., x_n) = \max\{x_1, ..., x_n\}$$

which is the most-commonly studied setting of this problem. This setting, known as the maximum regret case, provides us with a worst-case analysis of the game.

For $n = 2$, the solution is given in the sector $x_1 \geq x_2$ as follows [15]:

$$u(x) = x_1 + \frac{1}{2\sqrt{2}} e^{\sqrt{2}(x_2 - x_1)}. \tag{1.0.2}$$

For $n = 3$, the solution is given in the sector $x_1 \geq x_2 \geq x_3$ as follows [15]:

$$u(x) = x_1 + \frac{1}{2\sqrt{2}} e^{\sqrt{2}(x_2 - x_1)} + \frac{1}{6\sqrt{2}} e^{\sqrt{2}(2x_3 - x_2 - x_1)}. \tag{1.0.3}$$

For $n = 4$, the solution is given in the sector $x_1 > x_2 > x_3 > x_4$ as follows [8]:

$$\begin{aligned} u(x) = x_1 &- \frac{\sqrt{2}}{4}\sinh(\sqrt{2}(x_1 - x_2)) \\ &+ \frac{\sqrt{2}}{2}\arctan\left(e^{\frac{x_4+x_3-x_2-x_1}{2}}\right) \cdot \\ &\cosh\left(\frac{x_4 - x_3 + x_2 - x_1}{2}\right)\cosh\left(\frac{-x_4 + x_3 + x_2 - x_1}{2}\right)\cosh\left(\frac{-x_4 - x_3 + x_2 + x_1}{2}\right) \\ &+ \frac{\sqrt{2}}{2}\operatorname{arctanh}\left(e^{\frac{x_4+x_3-x_2-x_1}{2}}\right) \cdot \\ &\sinh\left(\frac{x_4 - x_3 + x_2 - x_1}{2}\right)\sinh\left(\frac{-x_4 + x_3 + x_2 - x_1}{2}\right)\sinh\left(\frac{-x_4 - x_3 + x_2 + x_1}{2}\right). \end{aligned} \tag{1.0.4}$$

For $n = 5$, a solution to the PDE has not yet been found, but experimental evidence suggests that the COMB strategy, which was previously conjectured to be optimal, is not optimal [9].

It would be interesting (although it may be difficult) to find explicit solutions of this PDE for larger values of n. Numerically solving these PDEs in higher dimensions allows us to approximate the value functions and optimal strategies in these higher dimensions. This work develops the theory of prediction with expert advice in the GH case past what is known within the literature, by developing a numerical PDE solver that can be used for an arbitrary number of experts, and demonstrably runs for at least 10 experts. In future work, we also plan to use numerical analysis and Monte Carlo methods to solve a PDE that gives insight into prediction with history-dependent experts [36, 37].

Chapter 2

Background

We describe here the problem of prediction from expert advice with *mixed strategies* (i.e., random strategies) against an adversarial environment. The setting consists of n experts making predictions, a player who chooses at each step which expert to follow, and an adversarial environment which decides which experts gain or lose at each step of the game. The strategies are *mixed*, so both the player and the adversary choose probability distributions over their possible plays, and their actual plays are random variables drawn from those distributions. At the kth step of the game, the player chooses a probability distribution α_k over the experts $\mathcal{A}_n := \{1, 2, \ldots, n\}$ and the adversary chooses a probability distribution β_k over the binary sequences $\mathcal{B}_n := \{0, 1\}^n$. Random variables from these distributions $i_k \sim \alpha_k$ and $\mathbf{v}_k \sim \beta_k$ are drawn, and the player chooses to follow expert i_k and the market advances the experts corresponding to the positions of the ones in the binary vector $\mathbf{v}_k$.

The performance of the player is measured by their regret to each expert, which is the difference between the expert's gains and those of the player. We let $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ denote the regret vector, so x_i is the current regret with respect to expert i. Let us write the coordinates of $\mathbf{v}_k$ as $\mathbf{v}_k = (v_{k,1}, v_{k,2}, \ldots, v_{k,n})$. Then on the kth step the the player accumulates regret of $v_{k,j} - v_{k,i_k}$ with respect to expert j. If the regret vector started at $x \in \mathbb{R}^n$ on the first step of the game, then the regret after T steps is

$$R_T := x + \sum_{k=1}^{T} (\mathbf{v}_k - v_{k,i_k} \mathbb{1})$$

At the end of the game the regret R_T is evaluated with a payoff function $g : \mathbb{R}^n \to \mathbb{R}$. The player's goal is to minimize $g(R_T)$ while the adversary's goal is to maximize $g(R_T)$. The most commonly used payoff is the maximum regret $g(x) = \max_{x_1, x_2, \ldots, x_n}$, which simply

reports the regret of the player with respect to the best performing expert. There are two common ways to stop the game: either it is played for a fixed number of steps T—the finite time horizon problem—or the game ends with probability $\delta > 0$ on each step—the geometric stopping problem. The geometric stopping problem corresponds to taking T to be a geometric random variable with success probability δ. Our numerical results in this paper are focused on the geometric stopping time problem.

For the geometric stopping time problem, the value function $U_\delta : \mathbb{R}^n \to \mathbb{R}$ is defined by

$$U_\delta(x) = \inf_\alpha \sup_\beta \mathbb{E}\left[g\left(x + \sum_{k=1}^{T} (\mathbf{v}_k - v_{k,i_k}\mathbb{1}) \right) \right], \tag{2.0.1}$$

where $\alpha = (\alpha_1, \alpha_2, \dots)$ and $\beta = (\beta_1, \beta_2, \dots)$. The value function $U_\delta(x)$ is the expected value of the payoff at the end of the game given that the regret vectors starts at $x \in \mathbb{R}^n$ on the first step and both entities play optimally. The inf and the sup are over *strategies for the players ('Entities' better than 'players'?*, which enforce that α_k and β_k depend only on the current value of the regret and the past choices of both players ('entities'?). The value function is unchanged by swapping the inf and sup in (2.0.1).

It was shown in [15, 16] that the rescaled value functions

$$u_\delta(x) := \sqrt{\delta} U_\delta\left(\frac{x}{\delta}\right)$$

converge locally uniformly, as $\delta \to 0$, to the viscosity solution of the degenerate elliptic partial differential equation (PDE)

$$u - \frac{1}{2} \max_{\mathbf{v} \in \{0,1\}^n} \mathbf{v}^T \nabla^2 u \, \mathbf{v} = g \quad \text{on} \quad \mathbb{R}^n. \tag{2.0.2}$$

The PDE (2.0.2) contains all the information about the prediction problem in the asymptotic regime where the number of steps T tends to ∞, which is equivalent to sending the geometric stopping probability δ to 0. It was shown in [16] that the asymptotically optimal player strategy is to use the probability distribution

$$\alpha_k = \nabla u(x), \tag{2.0.3}$$

where $x \in \mathbb{R}^n$ is the current regret on the kth step of the game, and the optimal adversary strategy is to choose

$$\mathbf{v}_k \in \operatorname*{argmax}_{\mathbf{v} \in \mathcal{B}_n} \{\mathbf{v}^T \nabla^2 u(x) \mathbf{v}\}, \tag{2.0.4}$$

and then advance the experts in $\mathbf{v}_k$ with probability $\frac{1}{2}$ and those in $\mathbb{1} - \mathbf{v}_k$ with the same probability.

Determining the adversary's strategies in the max-regret setting, where we take $g(x) = \max\{x_1, \dots, x_n\}$, is an open problem. It was suggested in [5] that the COMB strategy of ordering the experts by regret and using alternating zeros and ones $\mathbf{v} = (0, 1, 0, 1, \dots)$ is optimal in general, though this was only proven for $n = 2$ and $n = 3$ experts. The idea behind the COMB strategy is to group the experts into as equally matched groups as possible and advance one or the other with equal probability. However, the even numbered experts are slightly worse on average than compared to the odd numbered experts, so there is reason to believe that COMB can be improved upon for larger numbers of experts. It was shown in [8] that COMB is optimal for $n = 4$ as well. In each of the cases $n = 2, 3, 4$ there are other optimal strategies besides COMB. The question of whether COMB is optimal for $n \geq 5$ is an open problem. Some recent work [9] gives experimental evidence that COMB is *not optimal* for $n = 5$. It was shown in [33, 34] that COMB is at least as powerful as the setting of randomly choosing which half of the experts to advance (the Bernoulli strategy).

Chapter 3

Analysis of a General Numerical Scheme

3.1 Analysis of a general numerical scheme

We study here a finite difference scheme for the PDE (2.0.2) consisting of replacing the pure second derivatives $\mathbf{v}^T\nabla^2 u\mathbf{v}$ with finite difference approximations. We work on the grid $\mathbb{Z}_h^n$, where $\mathbb{Z}_h = h\mathbb{Z}$ and $h > 0$ is the grid spacing. For a function $u : \mathbb{Z}_h^n \to \mathbb{R}$, we define the discrete gradient as the mapping $\nabla_h : \mathbb{Z}_h^n \times \mathbb{Z}^n \to \mathbb{R}$ defined by

$$\nabla_h u(x, \mathbf{v}) := \frac{u(x + h\mathbf{v}) - u(x)}{h}. \tag{3.1.1}$$

The discrete Hessian is defined as the mapping $\nabla_h^2 u : \mathbb{Z}_h^n \times \mathbb{Z}^n \to \mathbb{R}$ given by

$$\nabla_h^2 u(x, \mathbf{v}) := \frac{u(x + h\mathbf{v}) - 2u(x) + u(x - h\mathbf{v})}{h^2}. \tag{3.1.2}$$

We note that

$$\nabla_h^2 u(x, \mathbf{v}) = \frac{\nabla_h u(x, \mathbf{v}) + \nabla_h(x, -\mathbf{v})}{h}. \tag{3.1.3}$$

Also, the definition of the discrete Hessian leads immediately to the discrete Taylor-type expansions

$$\frac{1}{2}(u(x + h\mathbf{v}) + u(x - h\mathbf{v})) = u(x) + \frac{h^2}{2}\nabla_h^2 u(x, \mathbf{v}), \tag{3.1.4}$$

and

$$u(x + h\mathbf{v}) = u(x) - h\nabla_h(u, -\mathbf{v}) + h^2\nabla_h^2 u(x, \mathbf{v}). \tag{3.1.5}$$

Define $\mathcal{H}_n = \{X : \mathbb{Z}_n \to \mathbb{R}\}$. We will use the notation $\nabla_h u(x) \in \mathcal{H}_n$ and $\nabla_h^2 u(x) \in \mathcal{H}_n$ for the mappings $\mathbf{v} \mapsto \nabla_h u(x, \mathbf{v})$ and $\mathbf{v} \mapsto \nabla^2 u_h(x, \mathbf{v})$, respectively. For $u \in C^{k,1}(\mathbb{R}^n)$ with $k = 2$ or $k = 3$ and any $\mathbf{v} \in \mathbb{Z}^n$ we have via Taylor expansion that

$$\nabla_h^2 u(x, \mathbf{v}) = \mathbf{v}^T \nabla^2 u(x) \mathbf{v} + \mathcal{O}(|\mathbf{v}|^{k+1} h^{k-1}).$$

Our discrete approximation of (2.0.2) is given by

$$u(x) - \frac{1}{2} \max_{\mathbf{v} \in \mathcal{B}_n} \nabla_h^2 u(x, \mathbf{v}) = g(x) \quad \text{for } x \in \mathbb{Z}_h^n.$$

Since our methods are not specific to this equation, we will study a more general class of equations of the form

$$u - F(\nabla_h^2 u) = g \quad \text{on } \mathbb{Z}_h^n, \tag{3.1.6}$$

where $F : \mathcal{H}_n \to \mathbb{R}$. The PDE (2.0.2) is obtained by setting

$$F(X) = \frac{1}{2} \max_{\mathbf{v} \in \{0,1\}^n} X(\mathbf{v}). \tag{3.1.7}$$

We need to place a monotonicity assumption on F.

Definition 3.1.1. We say that $F : \mathcal{H}_n \to \mathbb{R}$ is *monotone* if for all $X, Y \in \mathcal{H}_n$ with $X \leq Y$ we have $F(X) \leq F(Y)$.

For $X \in \mathcal{H}_n$ we define

$$\|X\|_{N,\infty} = \max_{\substack{\mathbf{v} \in \mathbb{Z}^n \\ |\mathbf{v}|_\infty \leq N}} |X(\mathbf{v})|, \tag{3.1.8}$$

where $|\mathbf{v}|_\infty = \max_{1 \leq \in \leq n} |v_i|$. We need to place a condition on the width of the stencil F.

Definition 3.1.2. We say $F : \mathcal{H}_n \to \mathbb{R}$ has *width* N if there exists $C > 0$ such that for all $X, Y \in \mathcal{H}_n$ we have

$$|F(X) - F(Y)| \leq C \|X - Y\|_{N,\infty}.$$

The smallest such constant $C > 0$ is called the *Lipschitz constant* of F and denoted $\mathrm{Lip}_N(F)$.

The choice of F given in (3.1.7) is monotone and has width $N = 1$, with $\mathrm{Lip}_1(F) = \frac{1}{2}$.

3.1.1 Existence and uniqueness

We first establish a comparison principle.

Theorem 3.1.3. *Assume F is monotone and has width N. Let $u, v : \mathbb{Z}_h^n \to \mathbb{R}$ satisfy*

$$u - F(\nabla_h^2 u) \leq v - F(\nabla_h^2 v) \quad \text{on } \mathbb{Z}_h^n, \tag{3.1.9}$$

and

$$\lim_{|x|\to\infty} \frac{u(x) - v(x)}{|x|^2} = 0. \tag{3.1.10}$$

Then $u \leq v$ on $\mathbb{Z}_h^n$.

Proof. For $\varepsilon > 0$ we define

$$u_\varepsilon(x) = u(x) - \frac{\varepsilon}{2}|x|^2 - \varepsilon N^2 - \varepsilon.$$

We claim that $u_\varepsilon \leq v$ for all $\varepsilon > 0$, from which the result follows. Fix $\varepsilon > 0$ and assume by way of contradiction that $\sup_{\mathbb{Z}_h^n}(u_\varepsilon - v) > 0$. By (3.1.10), there exists $R > 0$, depending on $\varepsilon > 0$, such that $u_\varepsilon(x) < v(x)$ for $|x| > R$. Thus, $u_\varepsilon - v$ attains its maximum over $\mathbb{Z}_h^n$ at some $x_0 \in \mathbb{Z}_h^n$, and $u_\varepsilon(x_0) > v(x_0)$. Since $u_\varepsilon(x) - v(x) \leq u_\varepsilon(x_0) - v(x_0)$ for all $x \in \mathbb{Z}_h^n$ we have

$$u_\varepsilon(x) - u_\varepsilon(x_0) \leq v(x) - v(x_0).$$

It follows that

$$\begin{aligned}\nabla_h^2 u_\varepsilon(x_0, \mathbf{v}) &= \frac{1}{h^2}(u_\varepsilon(x_0 + h\mathbf{v}) - u(x_0) + u_\varepsilon(x_0 - h\mathbf{v}) - u(x_0)) \\ &\leq \frac{1}{h^2}(v(x_0 + h\mathbf{v}) - u(x_0) + v(x_0 - h\mathbf{v}) - v(x_0)) = \nabla_h^2 v(x_0, \mathbf{v})\end{aligned}$$

for all directions $\mathbf{v} \in \mathbb{Z}^n$. Since F is monotone we have

$$F(\nabla_h^2 u_\varepsilon(x_0)) \leq F(\nabla_h^2 v(x_0)) \tag{3.1.11}$$

We now compute

$$\nabla_h^2 u_\varepsilon(x, \mathbf{v}) = \nabla_h^2 u(x, \mathbf{v}) - \varepsilon|\mathbf{v}|^2,$$

from which it follows that

$$\begin{aligned} u_\varepsilon(x_0) - F(\nabla_h^2 u_\varepsilon(x_0)) &= u(x_0) - \frac{\varepsilon}{2}|x_0|^2 - \varepsilon N^2 - \varepsilon - F(\nabla_h^2 u(x_0) - \varepsilon|\mathbf{v}|^2) \\ &\le u(x_0) - F(\nabla_h^2 u(x_0)) - \frac{\varepsilon}{2}|x_0|^2 - \varepsilon \\ &< v(x_0) - F(\nabla_h^2 v(x_0)) \\ &\le v(x_0) - F(\nabla_h^2 u_\varepsilon(x_0)), \end{aligned}$$

where the last line follows from (3.1.11). Therefore $u_\varepsilon(x_0) \le v(x_0)$, which is a contradiction. □

Existence of a solution follows from the comparison principle and the Perron method.

Theorem 3.1.4. *Assume F is monotone, has width N, and $F(0) = 0$. Suppose there exists $C_g > 0$ so that*

$$|g(x)| \le C_g(1 + |x|) \quad \textit{for all } x \in \mathbb{Z}_h^n. \tag{3.1.12}$$

Then there exists a unique solution $u : \mathbb{Z}_h^n \to \mathbb{R}$ of (3.1.6) satisfying $\lim_{|x|\to\infty} \frac{u(x)}{|x|^2} = 0$. Furthermore, we have

$$|u(x)| \le CC_g(\mathrm{Lip}_N(F)N^3 + 1 + |x|) \quad \textit{for all } x \in \mathbb{Z}_h^n, \tag{3.1.13}$$

where C depends only on n.

Proof. We define

$$\psi(x) = \sqrt{1 + |x|^2},$$

and note that ψ is a smooth function with linear growth satisfying

$$\frac{1}{\sqrt{2}}(1 + |x|) \le \psi(x) \le 1 + |x| \quad \text{for all } x \in \mathbb{R}^n. \tag{3.1.14}$$

For any $\mathbf{v} \in \mathbb{Z}^n$ and $x \in \mathbb{Z}_h^n$ we have

$$\begin{aligned} |\nabla_h^2 \psi(x, \mathbf{v})| &\le |\nabla_h^2 \psi(x, \mathbf{v}) - \mathbf{v}^T \nabla^2 \psi(x)\mathbf{v}| + C|\mathbf{v}|^2 \\ &\le |\nabla_h^2 \psi(x, \mathbf{v}) - \mathbf{v}^T \nabla^2 \psi(x)\mathbf{v}| + C|\mathbf{v}|^2 \\ &\le C\left(|\mathbf{v}|^3 h + |\mathbf{v}|^2\right) \le C|\mathbf{v}|^3, \end{aligned}$$

since $h \leq 1$ and $|\mathbf{v}| \geq 1$. Since $F(0) = 0$ we have

$$|F(\nabla_h^2 \psi)| = |F(\nabla_h^2 \psi) - F(0)| \leq \mathrm{Lip}_N(F) \|\nabla_h^2 \psi\|_{N,\infty} \leq C \,\mathrm{Lip}_N(F) N^3 =: \xi. \tag{3.1.15}$$

We now define

$$w = C_g(\xi + \sqrt{2}\psi).$$

Then by (3.1.14) and (3.1.15) we have

$$w - F(\nabla_h^2 w) = C_g \xi + \sqrt{2} C_g \psi - F(C_g \nabla_h^2 \psi) \geq \sqrt{2} C_g \psi \geq C_g(1 + |x|) \geq g.$$

A similar argument shows that $v = -w$ satisfies $v - F(\nabla_h^2 v) \leq g$.

Define

$$\mathcal{F} = \left\{ v : \mathbb{Z}_h^n \to \mathbb{R} \,:\, v - F(\nabla_h^2 v) \leq g \text{ and } v \leq w \right\} \tag{3.1.16}$$

and

$$u(x) = \sup\{v(x) \,:\, v \in \mathcal{F}\}. \tag{3.1.17}$$

Since $v = -w$ belongs to $\mathcal{F}$, the set $\mathcal{F}$ is nonempty and we have $-w \leq u \leq w$. Hence, u satisfies (3.1.13).

We now claim that

$$u - F(\nabla_h^2 u) \leq g \quad \text{on } \mathbb{Z}_h.$$

To see this, fix $x_0 \in \mathbb{Z}_h^n$ and let $v_k \in \mathcal{F}$ such that $\lim_{k\to\infty} v_k(x_0) = u(x_0)$. By passing to a subsequence, if necessary, we can assume that $\lim_{k\to\infty} v_k(x)$ exists for all $x \in \mathbb{Z}_h^n$. Let us denote $v(x) := \lim_{k\to\infty} v_k(x)$, noting that $v(x_0) = u(x_0)$. By continuity we have $v - F(\nabla_h^2 v) \leq g$ and $v \leq w$, thus $v \in \mathcal{F}$ and $v \leq u$. Since $v(x_0) = u(x_0)$, we have that $v - u$ attains its maximum at x_0. As in the proof of Theorem 3.1.3 we have $\nabla_h^2 v(x_0, \mathbf{v}) \leq \nabla_h^2 u(x_0, \mathbf{v})$ for all $\mathbf{v}$, and so by the monotonicity of F we have $F(\nabla_h^2 v(x_0)) \leq F(\nabla_h^2 u(x_0))$, which when combined with $u(x_0) = v(x_0)$ and $v - F(\nabla_h^2 v) \leq g$, establishes the claim.

To complete the proof, we show that

$$u - F(\nabla_h^2 u) \geq g \quad \text{on } \mathbb{Z}_h^n.$$

Assume to the contrary that there is some $x_0 \in \mathbb{Z}_h^n$ such that

$$u(x_0) - F(\nabla_h^2 u(x_0)) < g(x_0).$$

Define

$$v(x) = \begin{cases} u(x_0) + \varepsilon, & \text{if } x = x_0 \\ u(x), & \text{otherwise.} \end{cases}$$

By continuity we can choose $\varepsilon > 0$ small enough so that

$$u(x_0) - F(\nabla_h^2 u(x_0)) \leq g(x_0).$$

For $x \neq x_0$, we already have

$$u(x) - F(\nabla_h^2 u(x)) \leq g(x),$$

and the definition of v and the monotonicity of F imply that $\nabla^2 v(x, \mathbf{v}) \geq \nabla_h^2 u(x, \mathbf{v})$ for all $x \neq x_0$ and $\mathbf{v} \in \mathbb{Z}^n$. This completes the proof. □

3.1.2 Properties of solutions

We recall that a function $u : \mathbb{Z}_h^n \to \mathbb{R}$ is *Lipschitz continuous* if there exists $C > 0$ such that

$$|u(x) - u(y)| \leq C\|x - y\|$$

holds for all $x, y \in \mathbb{Z}_h^n$. The *Lipschitz constant* of u, denoted $\mathrm{Lip}(u)$, is the smallest such constant, given by

$$\mathrm{Lip}(u) = \sup_{\substack{x,y \in \mathbb{Z}_h^n \\ x \neq y}} \frac{|u(x) - u(y)|}{\|x - y\|}.$$

Lemma 3.1.5 (Basic properties)**.** *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Assume g satisfies (3.1.12) and let $u : \mathbb{Z}_h^n \to \mathbb{R}$ be the unique solution of (3.1.6). The following hold.*

(i) *If g is Lipschitz then so is u, and $\mathrm{Lip}(u) \leq \mathrm{Lip}(g)$.*

(ii) *If $F \geq 0$ then $u \geq g$ on $\mathbb{Z}_h^n$.*

(iii) *There exists a constant $C > 0$ such that*

$$\|u - g\|_\infty \leq C\,\mathrm{Lip}(g)\left(N\sqrt{\mathrm{Lip}_N(F)} + h\right). \tag{3.1.18}$$

Proof. (i) Let $z \in \mathbb{Z}_h^n$ and define $w(x) = u(x+z) - \mathrm{Lip}(g)\|z\|$. Then we have

$$w(x) - F(\nabla_h^2 w(x)) = u(x+z) - \mathrm{Lip}(g)\|z\| - F(\nabla_h^2 u(x+z)) = g(x+z) - \mathrm{Lip}(g) \leq g(x).$$

By the comparison principle, Theorem 3.1.3, we have $w \leq u$, and so

$$u(x+z) - u(x) \leq \mathrm{Lip}(g)\|z\|.$$

Since this holds for all $x, z \in \mathbb{Z}_h^n$, the proof of (i) is complete.

(ii) If $F \geq 0$ then $u(x) = g(x) + F(\nabla_h^2 u(x)) \geq g(x)$.

(iii) Let $\overline{g} : \mathbb{R}^n \to \mathbb{R}$ be the piecewise constant extension of g to a function on $\mathbb{R}^n$, defined so that $\overline{g}(y) = g(x)$ for all $y \in x + [0,h)^n$ and any $x \in Zh$. While the extension is discontinuous, it satisfies

$$|\nabla_h \overline{g}(y, \mathbf{v})| = \left| \frac{g(x + h\mathbf{v}) - g(x)}{h} \right| \leq \mathrm{Lip}(g)|\mathbf{v}|$$

provided $\mathbf{v} \in \mathbb{Z}^n$, where $x \in \mathbb{Z}_h^n$ satisfies $y \in x + [0,h)^n$. Let $\varepsilon > 0$ and define the standard mollification $g_\varepsilon := \eta_\varepsilon * \overline{g}$, where $\eta_\varepsilon(x) = \frac{1}{\varepsilon^d}\eta\left(\frac{x}{\varepsilon}\right)$, and η_ε is compactly supported in $B(0,\varepsilon)$. Note that

$$\nabla_h g_\varepsilon(x, \mathbf{v}) = \int_{\mathbb{R}^n} \eta_\varepsilon(y) \nabla_h \overline{g}(x-y, \mathbf{v})\, dy = \int_{\mathbb{R}^n} \eta_\varepsilon(x-z) \nabla_h \overline{g}(z, \mathbf{v})\, dz,$$

where we used the change of variables $z = x - y$ above. We also have

$$\nabla_h g_\varepsilon(x, -\mathbf{v}) = \int_{\mathbb{R}^n} \eta_\varepsilon(y) \nabla_h \overline{g}(x-y, -\mathbf{v})\, dy = -\int_{\mathbb{R}^n} \eta_\varepsilon(x-z-h\mathbf{v}) \nabla_h \overline{g}(z, \mathbf{v})\, dz,$$

where we now use the change of variables $z = x - y - h\mathbf{v}$. Therefore

$$\begin{aligned} \nabla_h^2 g_\varepsilon(x, \mathbf{v}) &= \frac{1}{h}(\nabla_h \overline{g}_\varepsilon(x, \mathbf{v}) + \nabla_h \overline{g}_\varepsilon(x, -v)) \\ &= -\int_{\mathbb{R}^n} \nabla_h \eta_\varepsilon(x-z, -\mathbf{v}) \nabla_h \overline{g}(z, \mathbf{v})\, dz. \end{aligned}$$

Therefore

$$|\nabla_h^2 g_\varepsilon(x, \mathbf{v})| \leq \int_{B(x,\varepsilon) \cup B(x-h\mathbf{v},\varepsilon)} |\nabla_h \eta_\varepsilon(x-z, -\mathbf{v})||\nabla_h \overline{g}(z, \mathbf{v})|\, dz \leq \frac{C}{\varepsilon} \mathrm{Lip}(g)|\mathbf{v}|^2.$$

Since F has width N, it follows that

$$|F(\nabla_h^2 g_\varepsilon(x))| = |F(\nabla_h^2 g_\varepsilon(x)) - F(0)| \leq \mathrm{Lip}_N(F)\|\nabla_h^2 g_\varepsilon\|_{N,\varepsilon} \leq \frac{C}{\varepsilon}\,\mathrm{Lip}_N(F)\,\mathrm{Lip}(g)N^2. \tag{3.1.19}$$

We also have that $|\overline{g}(y) - g(x)| \leq |y - x| + \sqrt{d}h$, and so

$$\begin{aligned} |g_\varepsilon(x) - g(x)| &= \left| \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)(\overline{g}(y) - g(x))\,dy \right| \\ &\leq \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)|\overline{g}(y) - g(x)|\,dy \\ &\leq \mathrm{Lip}(g)(\varepsilon + \sqrt{d}h). \end{aligned}$$

Combining this with (3.1.19) we have

$$|g_\varepsilon(x) + F(\nabla^2 g_\varepsilon(x)) - g(x)| \leq \mathrm{Lip}(g)(\varepsilon + \sqrt{d}h) + \frac{C}{\varepsilon}\,\mathrm{Lip}_N(F)\,\mathrm{Lip}(g)N^2.$$

Choosing $\varepsilon = N\sqrt{\mathrm{Lip}_N(F)}$ yields

$$|g_\varepsilon(x) + F(\nabla^2 g_\varepsilon(x)) - g(x)| \leq C\,\mathrm{Lip}(g)\left(N\sqrt{\mathrm{Lip}_N(F)} + h\right).$$

By the comparison principle, Theorem 3.1.3, we have

$$g_\varepsilon - C\,\mathrm{Lip}(g)\left(N\sqrt{\mathrm{Lip}_N(F)} + h\right) \leq u \leq g_\varepsilon + C\,\mathrm{Lip}(g)\left(N\sqrt{\mathrm{Lip}_N(F)} + h\right),$$

which completes the proof. □

To study further properties of u, we need some additional definitions.

Definition 3.1.6. We say that $F : \mathcal{H}_n \to \mathbb{R}$ is *convex* if for all $X, Y \in \mathcal{H}_n$ and $\lambda \in [0,1]$ we have

$$F(\lambda X + (1-\lambda)Y) \leq \lambda F(X) + (1-\lambda)F(Y).$$

Definition 3.1.7. We say that $u : \mathbb{Z}_h^n \to \mathbb{R}$ is *convex* if $\nabla_h^2 u(x) \geq 0$ for all $x \in \mathbb{Z}_h^n$.

We note that by (3.1.4), the convexity of u is equivalent to the inequality

$$u(x) \leq \frac{1}{2}(u(x + h\mathbf{v}) + u(x - h\mathbf{v}))$$

holding for all $x \in \mathbb{Z}_h^n$ and $\mathbf{v} \in \mathbb{Z}^n$. We also note that the choice of F given in (3.1.7) is convex.

We also consider a permutation invariance property.

Definition 3.1.8. We say $u : \mathbb{Z}_h^n \to \mathbb{R}$ is *permutation invariant* if $u \circ \sigma = u$ for all permutations σ on $\{1, \ldots, n\}$.

Definition 3.1.9. We say $F : \mathcal{H}_n \to \mathbb{R}$ is *permutation invariant* if $F(X \circ \sigma) = F(X)$ for all $X \in \mathcal{H}_n$ and all permutations σ on $\{1, \ldots, n\}$.

We note that F given in (3.1.7) is permutation invariant.

Finally, we also study translation properties.

Definition 3.1.10. For $\mathbf{v} \in \mathbb{Z}^n$, we say $u : \mathbb{Z}_h^n \to \mathbb{R}$ satisfies the $\mathbf{v}$-*translation property* if there exists a constant $c_{\mathbf{v}}$ such that

$$u(x + s\mathbf{v}) = u(x) + c_{\mathbf{v}} s \tag{3.1.20}$$

for all $x \in \mathbb{Z}_h^n$ and $s \in \mathbb{R}$ such that $s\mathbf{v} \in \mathbb{Z}_h^n$.

The next lemma shows that these properties for u are inherited from F and g.

Lemma 3.1.11. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Assume g satisfies (3.1.12) and let $u : \mathbb{Z}_h^n \to \mathbb{R}$ denote the unique solution of (3.1.6). The following hold:*

(i) *If F and g are convex, then u is convex.*

(ii) *If F and g are permutation invariant, then u is permutation invariant.*

(iii) *If g satisfies the $\mathbf{v}$-translation property with constant $c_{\mathbf{v}}$, then u does as well.*

Proof. (i) Let $\mathbf{v} \in \mathbb{Z}^n$ and define

$$w(x) = \frac{1}{2}(u(x + h\mathbf{v}) + u(x - h\mathbf{v})).$$

Then for $x \in \mathbb{Z}_h^n$ we compute, using the convexity of F, that

$$\begin{aligned}
w(x) - F(\nabla_h^2 w(x)) &= \frac{1}{2}u(x+h\mathbf{v}) + \frac{1}{2}u(x-h\mathbf{v}) - F\left(\frac{1}{2}\nabla_h^2 u(x+h\mathbf{v}) + \frac{1}{2}\nabla_h^2 u(x-h\mathbf{v})\right) \\
&\geq \frac{1}{2}u(x+h\mathbf{v}) + \frac{1}{2}u(x-h\mathbf{v}) - \frac{1}{2}F(\nabla_h^2 u(x+h\mathbf{v})) - \frac{1}{2}F(\nabla_h^2 u(x-h\mathbf{v})) \\
&= \frac{1}{2}g(x+h\mathbf{v}) + \frac{1}{2}g(x-h\mathbf{v}) \\
&= \frac{h^2}{2}\nabla_h^2 g(x,\mathbf{v}) + g(x) \\
&\geq g(x),
\end{aligned}$$

where the last line follows from the convexity of g. By Theorem 3.1.3 we have $w \geq u$, and so

$$\nabla_h^2 u(x,\mathbf{v}) = \frac{2}{h^2}(w(x) - u(x)) \geq 0$$

for all $x \in \mathbb{Z}_h^n$. Since $\mathbf{v} \in \mathbb{Z}^n$ is arbitrary, we have $\nabla_h^2 u(x) \geq 0$ for all $x \in \mathbb{Z}^n$, hence u is convex.

(ii) Let σ be a permutation and set $w = u \circ \sigma$. Then we have

$$w(x) - F(\nabla_h^2 w(x)) = u(\sigma(x)) - F(\nabla_h^2 (u \circ \sigma)(x)).$$

We note that for any $\mathbf{v} \in \mathbb{Z}^n$ we have $\nabla_h^2 (u \circ \sigma)(x,\mathbf{v}) = \nabla_h^2 u(\sigma(x), \sigma(\mathbf{v}))$. Therefore $\nabla_h^2 (u \circ \sigma)(x) = \nabla_h^2 u(\sigma(x)) \circ \sigma$. Since F is permutation invariant we have

$$F(\nabla_h^2 (u \circ \sigma)(x)) = F(\nabla_h^2 u(\sigma(x)) \circ \sigma) = F(\nabla_h^2 u(\sigma(x))).$$

Therefore

$$w(x) - F(\nabla_h^2 w(x)) = u(\sigma(x)) - F(\nabla_h^2 u(\sigma(x))) = 0.$$

By uniqueness of the solution u of (3.1.6) we have $w = u$, which completes the proof.

(iii) Let $s \in \mathbb{R}$ such that $s\mathbf{v} \in \mathbb{Z}_h^n$, and define $w(x) = u(x + s\mathbf{v}) - c_{\mathbf{v}} s$. Then

$$w(x) - F(\nabla_h^2 w(x)) = u(x+s\mathbf{v}) - c_{\mathbf{v}} s - F(\nabla_h^2 u(x+s\mathbf{v})) = g(x+s\mathbf{v}) - c_{\mathbf{v}} s = g(x).$$

Since the solution of (3.1.6) is unique, we have $w = u$, which completes the proof. □

3.1.3 Convergence rates

In this section, we prove convergence of the numerical scheme (3.1.6) towards the viscosity solution of the second order degenerate elliptic equation

$$u - F(\nabla^2 u) = g \quad \text{on} \quad \mathbb{R}^n \tag{3.1.21}$$

As before, we assume $F : \mathcal{H}_n \to \mathbb{R}$, and we interpret $F(X)$ for an $n \times n$ symmetric matrix X as

$$F(X) := F(\mathbf{v} \mapsto \mathbf{v}^T X \mathbf{v}).$$

We recall the definitions of viscosity solutions in Appendix A.1. Throughout this section we will use the notation $\mathrm{USC}(\mathcal{O})$ (resp. $\mathrm{LSC}(\mathcal{O})$) for the set of functions that are upper (resp. lower) semicontinuous at all points in $\mathcal{O} \subset \mathbb{R}^n$. For more details on the theory of viscosity solutions, we refer the reader to the user's guide [38] and [39].

Existence and uniqueness of a linear growth viscosity solution to (3.1.21) is standard material for viscosity solutions, and proofs can be found in [38, 39]. For use later on, we recall the comparison principle for (3.1.21) in Lemma 3.1.12 below. The proof of this result is standard in viscosity solution theory; a self-contained proof can be found in [39, Lemma 12.17].

Lemma 3.1.12. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Let $u \in USC(\mathbb{R}^n)$ be a viscosity subsolution of (3.1.21) and let $v \in LSC(\mathbb{R}^n)$ be a viscosity supersolution of (3.1.21). If*

$$\lim_{|x|\to\infty} \frac{u(x) - v(x)}{|x|^2} = 0 \tag{3.1.22}$$

then $u \leq v$ on $\mathbb{R}^n$.

Using this comparison principle and the Perron method, we can prove existence of a linear growth solution (Theorem 3.1.13 below). The application of the Perron method is standard and a self-contained proof can be found in [39, Theorem 12.18].[1]

Theorem 3.1.13. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Assume g is Lipschitz continuous and there exists $C_g > 0$ such that*

$$|g(x)| \leq C_g(1 + |x|). \tag{3.1.23}$$

[1] In fact, the proofs do not require Lipschitzness; uniform continuity is sufficient.

Then there exists a unique viscosity solution $u \in C(\mathbb{R}^n)$ of (3.1.21) satisfying

$$\lim_{|x|\to\infty} \frac{u(x)}{|x|^2} = 0.$$

Furthermore, there exists $C > 0$ such that

$$|u(x)| \leq C(1 + |x|). \tag{3.1.24}$$

Convergence of the discrete scheme (3.1.6) to the PDE (3.1.21) is a standard result in viscosity solution theory. We state the theorem in the following result and briefly sketch the proof.

Theorem 3.1.14. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Assume $g : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz continuous and satisfies (3.1.23). Let u_h be the solution of (3.1.6) and let u be the viscosity solution of (3.1.6). Then u_h converges to u locally uniformly as $h \to 0$, that is for all $R > 0$ we have*

$$\lim_{h\to 0} \max_{\substack{x\in\mathbb{Z}_h^n \\ |x|\leq R}} |u_h(x) - u(x)| = 0. \tag{3.1.25}$$

Proof. By Lemma 3.1.5 (i) we have $\mathrm{Lip}(u_h) \leq \mathrm{Lip}(g)$. By the Arzelá-Ascoli Theorem there exists a Lipschitz continuous function $u : \mathbb{R}^n \to \mathbb{R}$ such that, upon passing to a subsequence u_{h_k}, we have

$$\lim_{k\to\infty} \max_{\substack{x\in\mathbb{Z}_{h_k}^n \\ |x|\leq R}} |u_{h_k}(x) - u(x)| = 0$$

for all $R > 0$. The proof will be completed by showing that u is a viscosity solution of (3.1.21). By uniqueness of viscosity solutions of (3.1.21), the whole sequence u_h converges locally uniformly to u.

We verify the subsolution property for u; the supersolution property is similar. Let $x_0 \in \mathbb{R}^n$ and $\varphi \in C^\infty(\mathbb{R}^n)$ such that $u - \varphi$ has a local maximum at x_0. Without loss of generality we can assume x_0 is a strict global maximum of $u - \varphi$. It follows that there exists $x_k \to x_0$ such that $u_{h_k} - \varphi$ attains its maximum over $\mathbb{Z}_{h_k}^n$ at x_k. Therefore $\nabla_h^2 u_{h_k}(x_k, \mathbf{v}) \leq \nabla_h^2 \varphi(x_k)$, and since F is monotone we have

$$u(x_k) - F(\nabla_{h_k}^2 \varphi(x_k)) \leq \varphi(x_k) - F(\nabla_h^2 u_{h_k}(x_k)) = u(x_k) - u_{h_k}(x_k) - g(x_k).$$

Sending $k \to \infty$ we obtain

$$u(x_0) - F(\nabla^2 \varphi(x_0)) \leq g(x_0),$$

which completes the proof. □

If we have additional regularity for the solution u of (3.1.21), we can prove convergence rates. We recall the $C^{k,1}$ seminorm of $u : \mathbb{R}^n \to \mathbb{R}$ is defined by

$$[u]_{C^{k,1}(\mathbb{R}^n)} = \sum_{1 \leq |\alpha| \leq k} \operatorname{Lip}(D^\alpha u).$$

Theorem 3.1.15. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Assume $g : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz continuous and satisfies (3.1.23). Let u_h be the solution of (3.1.6) and let u be the viscosity solution of (3.1.6). If $[u]_{C^{k,1}(\mathbb{R}^n)} < \infty$ for $k = 2$ or $k = 3$, then we have*

$$\|u - u_h\|_\infty \leq C \operatorname{Lip}_N(F) [u]_{C^{k,1}(\mathbb{R}^n)} N^{k+1} h^{k-1} \tag{3.1.26}$$

Proof. By Taylor expansion we have

$$|\nabla_h^2 u(x, \mathbf{v}) - \mathbf{v}^T \nabla^2 u(x) \mathbf{v}| \leq C [u]_{C^{k,1}(\mathbb{R}^n)} |\mathbf{v}|^{k+1} h^{k-1}.$$

Therefore

$$|F(\nabla_h^2 u(x)) - F(\nabla^2 u(x))| \leq C \operatorname{Lip}_N(F) [u]_{C^{k,1}(\mathbb{R}^n)} N^{k+1} h^{k-1} =: \varepsilon.$$

It follows that

$$u(x) - F(\nabla_h^2 u(x)) \leq u(x) - F(\nabla^2 u(x)) + \varepsilon = g(x) + \varepsilon \quad \text{for } x \in \mathbb{Z}_h^n.$$

By Theorem 3.1.3 we have $u \leq u_h + \varepsilon$ on $\mathbb{Z}_h^n$. The opposite inequality is obtained similarly. □

From the convergence results in Theorems 3.1.14 and 3.1.15, we immediately obtain that all the properties of the discrete solutions proved in Section 3.1.2 extend to the viscosity solution u.

Proposition 3.1.16. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Assume $g : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz continuous and satisfies (3.1.23). Let u be the viscosity solution of (3.1.6). Then the following hold.*

(i) *u is Lipschitz continuous and* $\mathrm{Lip}(u) \leq \mathrm{Lip}(g)$.

(ii) *If $F \geq 0$ then $u \geq g$ on $\mathbb{R}^n$.*

(iii) *There exists a constant $C > 0$ such that*

$$\|u - g\|_\infty \leq CN \,\mathrm{Lip}(g)\sqrt{\mathrm{Lip}_N(F)}. \tag{3.1.27}$$

(iv) *If F and g are convex, then u is convex.*

(v) *If F and g are permutation invariant, then u is permutation invariant.*

(vi) *If g satisfies the $\mathbf{v}$-translation property with constant $c_{\mathbf{v}}$, then u does as well.*

We note that in (iv), the notion of convexity for F is the same as in Definition 3.1.6, while for u it is the usual one for functions on $\mathbb{R}^n$, that is

$$u(\lambda x + (1-\lambda)y) \leq \lambda u(x) + (1-\lambda)u(y).$$

In (v), the definition of permutation invariance for $u : \mathbb{R}^n \to \mathbb{R}$ is identical to Definition 3.1.8; namely $u = u \circ \sigma$. The translation property is defined similarly, but is slightly different so we give the definition below for functions on $\mathbb{R}^n$.

Definition 3.1.17. For $\mathbf{v} \in \mathbb{R}^n$, we say $u : \mathbb{R}^n \to \mathbb{R}$ satisfies the $\mathbf{v}$-*translation property* if there exists a constant $c_{\mathbf{v}}$ such that (3.1.20) holds for all $x \in \mathbb{R}^n$ and $s \in \mathbb{R}$.

The proof of Proposition 3.1.16 follows from Lemmas 3.1.5 and 3.1.11, and the convergence result in Theorem 3.1.14.

3.1.4 Restricting the domain

In order to compute the solution of the discrete scheme (3.1.6) on an unbounded domain $\mathbb{Z}_h^n$, it is necessary to restrict the domain to a compact set. In this section, we study restrictions of (2.0.2) to *computational* domains of the form $\Omega_{T,h} := [-T, T]^n \cap \mathbb{Z}_h^n$. In this section we always assume $T = mh$ for some integer $m \geq 1$. We define the width N boundary as

$$\partial_N \Omega_{T,h} = \{x \in \mathbb{Z}_h^n \setminus \Omega_{T,h} \,:\, \text{there exists } y \in \Omega_{T,h} \text{ with } |x - y|_\infty \leq N\}. \tag{3.1.28}$$

We set a Dirichlet boundary condition on $\partial_N \Omega_{T,h}$ and show that this condition affects the solution only near the boundary, and the solution remains accurate in the interior of $\Omega_{T,h}$.

In this section we study the equation

$$u - F(\nabla_h^2 u) = g \quad \text{on} \quad \Omega_{T,h}, \tag{3.1.29}$$

which is the restriction of (3.1.6) to the computational domains $\Omega_{T,h}$. We first recall the comparison principle for (3.1.29).

Lemma 3.1.18. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Let $u, v : \mathbb{Z}_h^n \to \mathbb{R}$ satisfy*

$$u - F(\nabla_h^2 u) \leq v - F(\nabla^2 v) \quad \text{on} \quad \Omega_{T,h} \tag{3.1.30}$$

and $u \leq v$ on $\partial_N \Omega_{T,h}$. Then $u \leq v$ on $\Omega_{T,h}$.

Proof. Let $x_0 \in \Omega_{T,h} \cup \partial_N \Omega_{T,h}$ be a point where $u - v$ attains its maximum value over $\Omega_{T,h} \cup \partial_N \Omega_{T,h}$. If $x_0 \in \partial_N \Omega_{T,h}$ then $u \leq v$, so we may assume $x_0 \in \Omega_{T,h}$. In this case we have $\nabla_h^2 u(x_0, \mathbf{v}) \leq \nabla_h^2 v(x_0, \mathbf{v})$, and since F is monotone, we obtain

$$u(x_0) - v(x_0) \leq F(\nabla_h^2 u(x_0)) - F(\nabla_h^2 v(x_0)) \leq 0.$$

This completes the proof. □

We now establish localization of solutions of (3.1.29).

Theorem 3.1.19. *Assume F is monotone, has width N, and satisfies $F(0) = 0$. Let $u, v : \mathbb{Z}_h^n \to \mathbb{R}$ satisfy (3.1.30). Then for any $\alpha \in (0, 1)$ we have*

$$\max_{\Omega_{\alpha T,h}} (u - v) \leq \frac{C \operatorname{Lip}_N(F) N^2}{(1-\alpha)^2 T^2} \left(\log \left(\frac{T^2}{\operatorname{Lip}_N(F) N^2} \right)^2 + 1 \right) \max_{\partial_N \Omega_{T,h}} (u - v). \tag{3.1.31}$$

Remark 3.1.20. The estimate (3.1.31) in Theorem 3.1.19 shows that the Dirichlet boundary conditions on $\partial_N \Omega_{T,h}$ have a limited domain of influence on the solution in $\Omega_{T,h}$ when $T > 0$ is large. Indeed, if we fix, say, $\alpha = \frac{1}{2}$, and if $u, v : \mathbb{Z}_h^n \to \mathbb{R}$ are solutions of (3.1.29), then it follows from Theorem 3.1.19 that

$$\max_{\Omega_{T/2,h}} |u - v| \leq \frac{C \log(T)^2}{T^2} \max_{\partial_N \Omega_{T,h}} |u - v|, \tag{3.1.32}$$

where C depends on $\operatorname{Lip}_N(F)$ and N, and $T \geq 3$. This shows that errors in the Dirichlet condition on $\partial_N \Omega_{T,h}$ can be tolerated numerically, provided $T > 0$ is large enough. For example, by Lemma 3.1.5 (iii), we can set $u = g$ on $\partial_N \Omega_{T,h}$ and obtain an $\mathcal{O}(\log(T)^2/T^2)$

approximation of the solution of (3.1.21) on the interior domain $\Omega_{T/2,h}$. We show how to obtain even more accurate solutions with better choices of boundary conditions in Section **??**.

Proof. Let $\mu = \max_{\partial_N \Omega_{T,h}}(u - v)$ and define

$$w(x) = v(x) + \gamma + \mu \sum_{i=1}^{n} \left(e^{\lambda\left(\frac{x_i}{T} - 1\right)} + e^{-\lambda\left(\frac{x_i}{T} + 1\right)} \right) \tag{3.1.33}$$

for parameters $\lambda, \gamma > 0$ to be determined. For $x \in \partial_N \Omega_{T,h}$, there exists i such that $x_i \geq T$ or $x_i \leq -T$, and so $w(x) \geq v(x) + \mu \geq u(x)$. Note that we have

$$|F(\nabla_h^2 w) - F(\nabla_h^2 v)| \leq \mathrm{Lip}_N(F) \|\nabla_h^2 w - \nabla_h^2 v\|_{N,\infty} \leq C \,\mathrm{Lip}_N(F) \frac{\mu \lambda^2 N^2}{T^2}.$$

Choosing $\gamma = C \,\mathrm{Lip}_N(F) \mu \lambda^2 N^2 T^{-2}$ we find that

$$w(x) - F(\nabla_h^2 w(x)) \geq v(x) + \gamma - F(\nabla_h^2 v(x)) - C \,\mathrm{Lip}_N(F) \mu \lambda^2 N^2 T^{-2} \geq 0.$$

By Lemma 3.1.18 we have $w \geq u$ on $\Omega_{T,h}$, and so

$$u(x) - v(x) \leq \gamma + \mu \sum_{i=1}^{n} \left(e^{\lambda\left(\frac{x_i}{T} - 1\right)} + e^{-\lambda\left(\frac{x_i}{T} + 1\right)} \right)$$

for all $x \in \Omega_{T,h}$. For $x \in \Omega_{\alpha T,h}$ with $\alpha \in (0,1)$ we have

$$u(x) - v(x) \leq \gamma + 2n\mu e^{-\lambda(1-\alpha)}.$$

Choosing λ so that $(1-\alpha)\lambda = \log\left(\frac{T^2}{\mathrm{Lip}_N(F) N^2}\right)$ completes the proof. $\square$

Chapter 4

Implementation of the Numerical Scheme

4.1 Implementation of the Numerical Scheme

We now return to a discussion of the elliptic PDE (2.0.2) that encodes information about the problem of prediction with expert advice with n experts. Section **??** describes that the discretization of (2.0.2) in the reduced domain (positive sector) can be given by

$$w(x) - \frac{1}{2} \max_{\mathbf{v} \in \mathcal{B}_{n-1}} \nabla_h^2 w(x, \mathbf{v}) = g(x) \quad \text{for } x \in \mathbb{D}_n^+.$$

The following sections describe a numerical scheme that is used to solve this equation, and elements of the design of the scheme that make the problem computationally tractable for larger values of n.

For $x \in \mathbb{D}_n^+$ and grid spacing h, we write $\alpha = h^{-1}x \in \mathbb{Z}^n$, and call α a multiindex. The relationship between the multiindex and the points in the computational domain is $x = h\alpha$, $y = h\beta$. The x and y are regret variables and α and β are corresponding multiindices for the numerical scheme. This simplifies notation in this section.

4.1.1 Numerical Scheme

We use a finite-difference scheme with the following definition:

$$F = \frac{1}{2} \max_{\mathbf{v} \in \mathcal{B}_{n-1}} \mathbf{v}^T \nabla_h^2 w \mathbf{v} + g - w \tag{4.1.1}$$

The procedure for the numerical scheme is as follows. Until the residual (defined as $|F|$)

is under a specified tolerance, repeat the following steps:

1. Impose the boundary conditions, discussed in Section 4.1.4.

2. Calculate F as in (4.1.1).

3. Update the estimate of w. This can either use Gradient Descent to calculate $w^{(t+1)}$ based on the estimate at the previous step $w^{(t)}$:

$$w^{(t+1)} = w^{(t)} + F\ dt \tag{4.1.2}$$

   or can use PDE Acceleration to calculate $w^{(t+1)}$ based on the previous two steps $w^{(t)}$ and $w^{(t-1)}$. PDE Acceleration [40] is similar to Nesterov/momentum methods, with a friction term a that can be hand-tuned for faster convergence. It provides a faster alternative to gradient descent, and is the method that was used for the results in this paper. This update is given by eq. (59) in [40]:

$$w^{(t+1)} = \frac{dt^2 F - w^{(t-1)} + (2 + dt\ a)w^{(t)}}{1 + a\ dt} \tag{4.1.3}$$

   Given the spatial grid spacing h, the time step size dt can be given by the following equation from section 3.1 of [40]:

$$dt = C\frac{\sqrt{2}h}{\sqrt{1+h^2}} \tag{4.1.4}$$

   for some constant $C < 1$. In the experiments in this paper, $C = 0.9$ was used.

   The PDE acceleration constant a can be tuned to give faster convergence. We tuned by trying a range of values and choosing the constant that led to convergence in the lowest number of iterations. We started with an equation for the constant given in [40]:

$$a = C\sqrt{1 + (T/\pi)^2}. \tag{4.1.5}$$

The analysis in [40] for figuring out an optimal damping constant is not directly applicable to this PDE, so we experimentally hand-tuned it to figure out the best value for C by observing which value led to the lowest number of iterations to convergence. Experimentally, this was $C = 2$, so the value $a = 2\sqrt{1 + (T/\pi)^2}$ is used for the results within this paper.

4.1.2 Domain Reduction

We describe some computational details of a domain reduction that allows for more tractable numerical solution of the n-expert geometric horizon problem.

By the results in Section 3.1.4, we are able to reduce the entire domain $\mathbb{R}^n$ to a bounded box $[0,T]^n$, giving us a bounded computational domain that we can discretize and use for numerics. However, this domain is of size m^n for a domain with $m = T/h$ grid points in each direction and n experts, which becomes unmanageable for fine grid resolutions for higher numbers of experts due to the curse of dimensionality in the number of experts n. To mitigate this, we can use the dimension-reduction result (Lemma **??**) to convert the problem of n experts in n dimensions to an equivalent problem in $d = n-1$ dimensions where $x_n = 0$, bringing the number of points to m^{n-1}. Finally, if we apply a reflection condition (Proposition **??**), we can further reduce to the sector

$$\mathbb{S}(T,n) = \{x : T \geq x_1 \geq x_2 \geq \cdots \geq x_{n-1} \geq x_n = 0.\}$$

This leads to a domain with a total number of gridpoints given by

$$\mathcal{N}(m,d) = \sum_{i_1=1}^{m} \sum_{i_2=1}^{i_1} \cdots \sum_{i_{d-1}=1}^{i_{d-2}} \sum_{i_d=1}^{i_{d-1}} 1$$

where $d = n-1$ is the dimension of the problem after the domain reduction, and m was the number of gridpoints in each direction.

Proposition 4.1.1. $\mathcal{N}(m,d) = \binom{m+d-1}{d}$.

Proof. To show this, we have that

$$\begin{aligned}
\mathcal{N}(m,d) &= \sum_{i_1=1}^{m} \cdots \sum_{i_{d-1}=1}^{i_{d-2}} \sum_{i_d=1}^{i_{d-1}} 1 \\
&= \sum_{i_1=1}^{m} \cdots \sum_{i_{d-1}=1}^{i_{d-2}} i_{d-1} \\
&= \sum_{i_1=1}^{m} \cdots \sum_{i_{d-2}=1}^{i_{d-3}} \binom{i_{d-2}+1}{2} \\
&= \sum_{i_1=1}^{m} \cdots \sum_{i_{d-3}=1}^{i_{d-4}} \binom{i_{d-3}+2}{3}.
\end{aligned}$$

At each summation, we apply the formula

$$\sum_{k=1}^{q} \binom{k+a-1}{a} = \binom{q+a}{a+1}$$

where $k = i_{d-b}$, $q = i_{d-(b+1)}$, and $a = b$. This leads to the simplification

$$\sum_{i_{d-b}=1}^{i_{d-(b+1)}} \binom{i_{d-b}+(b-1)}{b} = \binom{i_{d-(b+1)}+b}{b+1}.$$

This summation is done for $b = 1, \ldots, d-1$, with the final summation being

$$\sum_{i_{d-(d-1)}=1}^{m} \binom{i_{d-(d-1)}+((d-1)-1)}{d-1} = \binom{m+(d-1)}{(d-1)+1}$$

$$\sum_{i_1=1}^{m} \binom{i_1+(d-2)}{d-1} = \binom{m+(d-1)}{d}$$

so we can conclude that

$$\mathcal{N}(m,d) = \binom{m+d-1}{d}. \quad \square$$

$\square$

As an example, Table 4.1 gives the number of total gridpoints used in the numerical experiments (using these reductions), compared with the number of gridpoints that would have been necessary without the reductions.

We now describe how the curse of dimensionality in the number of gridpoints is resolved by the reduction of the domain from $[0,T]^n$ to sector(T,n). In particular, we describe the asymptotic behavior of $\mathcal{N}(m,d)$ as the number of gridpoints per direction in the original domain, m, is held constant and the dimension $d = n-1$ is sent to $d \to \infty$. Note that without the domain reduction, we run into the curse of dimensionality, since the total number of gridpoints in the domain is m^d (exponential in the dimension).

We can look at how $\mathcal{N}(m,d)$ scales as m is held constant and $d \to \infty$ through the following scaling lemma:

Lemma 4.1.2. *$\mathcal{N}(m,d) \sim C_m d^{m-1}$ as $d \to \infty$, with m constant, where C_m is a constant depending only on m.*

Proof. We will start with a simplification that will be used in the proof:

Recall that

$$\mathcal{N}(m,d) = \binom{m+d-1}{d} = \frac{(m+d-1)!}{d!(m-1)!}.$$

We will use Stirling's approximation ($n! \sim \sqrt{2\pi n}(\frac{n}{e})^n$) to simplify some of the factorial terms. Additionally, since we are just looking at the limit as $d \to \infty$, terms just depending on m are constant. We have that

$$\begin{aligned}
\mathcal{N}(m,d) &= \frac{(m+d-1)!}{d!(m-1)!} \\
&\approx \frac{\sqrt{2\pi(m+d-1)}(\frac{m+d-1}{e})^{m+d-1}}{\sqrt{2\pi d}(\frac{d}{e})^d(m-1)!} \\
&= \frac{e^{m-1}}{(m-1)!}\sqrt{\frac{m+d-1}{d}}\frac{(m+d-1)^{m+d-1}}{d^d}
\end{aligned}$$

The term

$$\sqrt{\frac{m+d-1}{d}} \to 1$$

as $d \to \infty$, and

$$\frac{e^{m-1}}{(m-1)!}$$

is constant, since it only depends on m.

For the next term, we have that

$$\frac{(m+d-1)^{m+d-1}}{d^d} \geq (\frac{m+d-1}{d})^d.$$

Using the above simplifications, this lemma is equivalent to showing that

$$\lim_{d\to\infty} \frac{\frac{(m+d-1)^{m+d-1}}{d^d}}{d^{m-1}} = C_m$$

where C_m is just a constant depending on m. Simplifying further, this would mean

$$\lim_{d\to\infty}(\frac{m+d-1}{d})^{m+d-1} = C_m$$

or alternately

$$\lim_{d\to\infty}(1+\frac{m-1}{d})^{m+d-1} = C_m.$$

To this end, we will look at $\lim_{d\to\infty}(1+\frac{m-1}{d})^{m+d-1}$ with m constant. To simplify the notation, we can define a new term M and set

$$\frac{1}{M}=\frac{m-1}{d}.$$

In doing so, sending $d\to\infty$ also corresponds to sending $M\to\infty$, since d and M only differ by a scaling factor, $d=M(m-1)$. This limit can then be rewritten as follows:

$$\begin{aligned}
&\lim_{M\to\infty}\left(1+\frac{1}{M}\right)^{m+d-1}\\
&=\lim_{M\to\infty}\left(1+\frac{1}{M}\right)^{m+M(m-1)-1}\\
&=\lim_{M\to\infty}\left(1+\frac{1}{M}\right)^{m-1}\left(1+\frac{1}{M}\right)^{M(m-1)}\\
&=\lim_{M\to\infty}\left(1+\frac{1}{M}\right)^{m-1}\left(\left(1+\frac{1}{M}\right)^{M}\right)^{m-1}.
\end{aligned}$$

For the first term in the product, $1+\frac{1}{M}\to 1$ as $M\to\infty$, so $\left(1+\frac{1}{M}\right)^{m-1}\to 1$ as $M\to\infty$.

For the second term, we know that $\left(1+\frac{1}{M}\right)^{M}\to e$ as $M\to\infty$, so $\left(\left(1+\frac{1}{M}\right)^{M}\right)^{m-1}\to e^{m-1}$ as $M\to\infty$.

Putting it all together, the given limit is $\frac{e^{2(m-1)}}{(m-1)!}$ as $d\to\infty$, with m constant.

Based on this, we have that $\mathcal{N}(m,d)\sim\frac{e^{2(m-1)}}{(m-1)!}d^{m-1}$ as $d\to\infty$, with m constant. □

Remark 4.1.3. We see from this that the growth is now polynomial in the dimension d, rather than exponential in d (as it was before these reductions of the domain, when the number of gridpoints was m^d). The reduction to the sector is crucial in avoiding the curse of dimensionality, and allows this problem to be solved numerically in higher dimensions (i.e. for higher numbers of experts).

As seen in Table 4.1, the reduction of the domain reduces the total number of gridpoints significantly by overcoming the curse of dimensionality in the number of gridpoints. However, one can note that the number of gridpoints that is manageable still decreases with increasing n. This is due to the curse of dimensionality in the operator itself, so the number of gridpoints overall must be decreased in order to keep the overall computational complexity manageable. Section 4.1.5 discusses some methods that can improve the computational complexity.

n	m	Total # gridpoints, $[0,T]^n$	Total # gridpoints, sector(T,n)
3	1000	1.0×10^9	5.0×10^5
4	250	3.9×10^9	2.6×10^6
5	150	7.6×10^{10}	2.2×10^7
6	65	7.5×10^{10}	1.1×10^7
7	35	6.4×10^{10}	3.8×10^6
8	25	1.5×10^{11}	2.6×10^6

Table 4.1: Total number of gridpoints in the original domain vs. the reduced domain.

4.1.3 Flattening the Domain

We describe the new indexing convention used within the code to solve the PDE strictly over the sector. Previously, for a d-dimensional problem, we have worked with the d-dimensional multiindex $\alpha = (\alpha_1, ..., \alpha_d)$ where $x = h\alpha$. Instead, when working over the sector, we work with a one-dimensional index (here also referred to as the flattened index). This allows us to only index over points in the computational domain (positive sector). The main difficulty is that the computational scheme needs efficient access to the neighbors in the sector when the sector is stored in a sparse (i.e., flattened) format. This section gives formulas for the flattened indices, and explains how to find neighbors, which is necessary to take derivatives and impose boundary conditions.

Consider the computation box grid $G_m = \{0, 1, \ldots, m-1\}^d$. We let a multiindex $\alpha \in G_m$ denote the index of a typical grid point. Let $\ell : G_m \to \mathbb{N}_0$ be the flattened index for the box, so

$$\ell(\alpha) = \sum_{k=1}^{d} \alpha_k m^{k-1}.$$

Define the sector

$$D_m = \{\alpha \in G_m \,:\, \alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_n \geq 0\}.$$

Note that $\mathcal{N}(m,d)$ is the total number of points in D_m. Let $\tau : D_m \to \mathbb{N}_0$ be an array of the linear indices $\ell(\alpha)$ for each $\alpha \in D_m$, that is, τ is the vector of length $\mathcal{N}(m,d)$ of linear indices corresponding to grid points in the sector in the order we visit them in the code

The point is that it is very easy computationally to compute $\ell(\alpha)$ or $\ell(\alpha+\beta)$ for any α, β. But our solution w is a length $N_{m,d}$ array, $w : D_m \to \mathbb{N}_0$, so we can't index w with $\ell(\alpha)$ directly. The observation is that if we want to index w at α, we just need to find

$1 \leq i \leq N_{m,d}$ such that

$$\tau(i) = \ell(\alpha).$$

When we flatten the sector, it becomes nontrivial to find neighboring gridpoints, which is necessary for imposing boundary conditions and calculating derivatives, so we need a way to find find i such that $\tau(i) = \ell(\alpha)$. To find i such that $\tau(i) = \ell(\alpha)$, we use the fact that τ is an increasing array, so we can use a bisection search. This takes $\log_2(N_{m,d})$ operations to find the corresponding index i. In other words, this is a sparse vector representation of D_m.

4.1.4 Boundary Conditions

In this section, we describe the implementation of the boundary conditions for the PDE 1.0.1, and formulas that we use to work within the modified domain (truncated to be finite, reduced by dimension, and reduced to the sector $x_1 \geq x_2 \geq \cdots \geq x_{n-1}$). We use the notation $w(x) = w[\alpha]$ to switch between notation in terms of the point and the multiindex.

To solve this PDE over the truncated domain, we have three main conditions to impose:

1. Large-x boundary condition: when $\alpha_i = m - 1$ for α_i an index in α (equivalently if $x_1 = T$), we set the approximate boundary condition

   $$w[\alpha] = g[\alpha]$$

   (In practice, g can be replaced with any other function that is within $O(1)$ of w: another good choice of boundary condition is the solution to the $d-1$-dimensional problem in cases where that is known.) This is an application of Theorem 3.1.19 and Remark 3.1.20.

2. Oblique boundary condition: when an element α_i of α has $\alpha_i = -1$, impose the condition

   $$w[\alpha] = w[\alpha + \beta] - h$$

   where $\beta = (b_1, ..., b_d)$ is a multiindex with $b_k = 1$ for $k \neq i$, $b_k = 2$ for $k = i$. This is an application of Lemma **??**.

3. Reflection condition: Heuristically, this means that if we want to calculate the value of $w[\alpha]$ for some α-value outside of the sector $\alpha_1 \geq \cdots \geq a_d$ (i.e., an unordered index), we can note that $w[\alpha] = w[\pi(\alpha)]$ for $\pi(\alpha)$ any permutation of the indices. Consequently, we can set

   $$w[\alpha] = w[\pi^*(\alpha)] \tag{4.1.6}$$

for $\pi^*(\alpha)$ the permutation that orders the elements of α in decreasing order, therefore pointing to an element within the sector. This is also an application of Lemma **??**.

4.1.5 Techniques for improving efficiency

We have already overcome the curse of dimensionality in the size of the computational domain, but the curse still appears in the number of directions in $\{0,1\}^n$ that we need to compute directional derivatives in. So we have overcome the *memory* curse of dimensionality, but not the *computational* curse. In this section we give some approximations aimed at overcoming the computational curse of dimensionality.

Although the reductions to the domain resolved the curse of dimensionality in the number of gridpoints, the curse of dimensionality still appears in terms of computational complexity, since we are taking the maximum over 2^n vectors.

The following methods replace the large set $\{0,1\}^n$ by a smaller set $V_t \subset \{0,1\}^n$ at each iteration t. The way this is chosen is different in each method. In the first, we choose $V_t = \{v_{t_k}\}$, where $t_k = t - (t \mod k)$ and $\mathbf{v}_t$ is optimal at time t. In the second, we shoose V_t to be a random subset of $\{0,1\}^n$ of much smaller size. The methods are outlined below:

1. Choose a constant k (ex: 5, 10, 20). Start by calculating all directional derivatives and figuring out the maximizing direction at each point. Then, keep iterating in that direction for k iterations before recalculating all derivatives, figuring out the new maximizing direction at each point, and iterating in the new directions k times. Repeat until convergence.

2. Choose the maximizing direction over a random subset of the directions.

Either of these options significantly decrease the runtime, since in the majority of the iterations, fewer derivatives are being calculated.

Justification of Random Direction Choice

In this section, we provide justification for the method of choosing the maximizing direction over a random subset of the directions $\mathbf{v} \in \mathcal{B}_n$. By choosing a smaller subset k of directions to take directional derivatives and compare, we can reduce the issues resulting from the curse of dimensionality in the operator F.

We introduce the notation $\Sigma_k \subset \mathcal{B}_{n-1}$ to refer to a subset of k randomly chosen vectors. Further, we introduce the following functions defined over vectors $\mathbf{v}$ in either the entire collection $\mathcal{B}_{n-1}$ or the subset Σ_k:

$$h_{\mathcal{B}_{n-1}} = \max_{\mathbf{v}\in\mathcal{B}_{n-1}} \mathbf{v}^T \nabla^2 w \mathbf{v} \tag{4.1.7}$$

$$h_{\Sigma_k} = \max_{\mathbf{v}\in\Sigma_k} \mathbf{v}^T \nabla^2 w \mathbf{v} \tag{4.1.8}$$

We can then look to quantify how well $\max_{\Sigma_k}$ approximates $\max_{\mathcal{B}_{n-1}}$. To do so, define the function

Lemma 4.1.4.

$$\mathbb{P}[h_{\mathcal{B}_{n-1}} - h_{\Sigma_k} = 0] \geq 1 - e^{-kp} \tag{4.1.9}$$

where p is the ratio of optimal strategies in $\mathcal{B}_n$.

Proof. We can start by noting that

$$\mathbb{P}[h_{\mathcal{B}_{n-1}} - h_{\Sigma_k} = 0] = 1 - \mathbb{P}[h_{\mathcal{B}_{n-1}} - h_{\Sigma_k} > 0].$$

$\mathbb{P}[h_{\mathcal{B}_{n-1}} - h_{\Sigma_k} > 0]$ is the probability of choosing k non-optimal strategies, so

$$\mathbb{P}[h_{\mathcal{B}_{n-1}} - h_{\Sigma_k} > 0] = (1-p)^k.$$

Therefore

$$\mathbb{P}[h_{\mathcal{B}_{n-1}} - h_{\Sigma_k} = 0] = 1 - (1-p)^k$$

$$\geq 1 - e^{-kp}.$$

□

By choosing k appropriately, we can therefore make the probability of having an optimal strategy high enough.

Chapter 5

Numerical Experiments

5.1 Numerical experiments

5.1.1 Evidence for Convergence of the Scheme

In order to test whether the numerical scheme converges to the correct solution, we used known solutions for the 3 and 4-expert problems ((1.0.3), (1.0.4)), as well as a different known solution, given in [33, 34], that the exact solution of

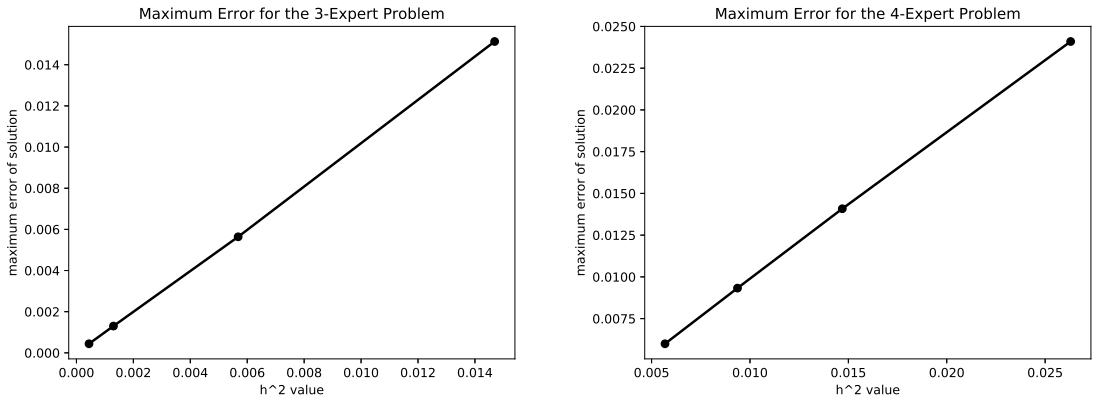
$$u - \frac{1}{2} \max_{\mathbf{v} \in \{e_i\}_{i=1}^n} \mathbf{v}^T \nabla^2 u \, \mathbf{v} = \max\{x_1, x_2, \ldots, x_n\} \quad \text{on} \quad \mathbb{R}^n \tag{5.1.1}$$

is given by

$$u(x) = x_1 + \sum_{k=1}^{n-1} \frac{e^{-\sqrt{2}(x_1 + \cdots + x_k - k x_{k+1})}}{\sqrt{2} k (k+1)}. \tag{5.1.2}$$

and measured the error between our scheme and the exact solution given in these results. We performed these experiments in two settings: (1) with boundary conditions given by the exact solution, and (2) with boundary conditions given by $g(x) = \max\{x_1, \ldots, x_d\}$.

Setting (1) allowed us to view whether the numerical solution would converge to the exact solution on the whole numerical domain, if given the correct boundary condition. Setting (2) allowed us to check if, given an approximate boundary condition at $x_1 = T$, whether the numerical solution would converge to the exact solution on some portion of the domain away from $x_1 = T$. Setting (2) was important in order to assess the reliability of solutions for $n \geq 5$ where we do not have exact solutions and must rely on an approximate boundary condition.



(a) Error vs. h^2, 3 experts

(b) Error vs. h^2, 4 experts

Figure 5.1: In both of these plots, we can see that maximum error scales linearly with h^2.

Convergence with Exact Boundary Conditions

We observe that, when the exact (known) boundary condition is imposed for the 3- and 4-expert problems, maximum error over the domain scales with order h^2.

Convergence with Approximate Boundary Conditions

In this section, we calculated the maximum error over the inner $K\%$ of the domain, defined as follows:

$$\text{inner } K\% \text{ of the domain } = \mathbb{S}(K \times 10^{-2}T, n) \tag{5.1.3}$$

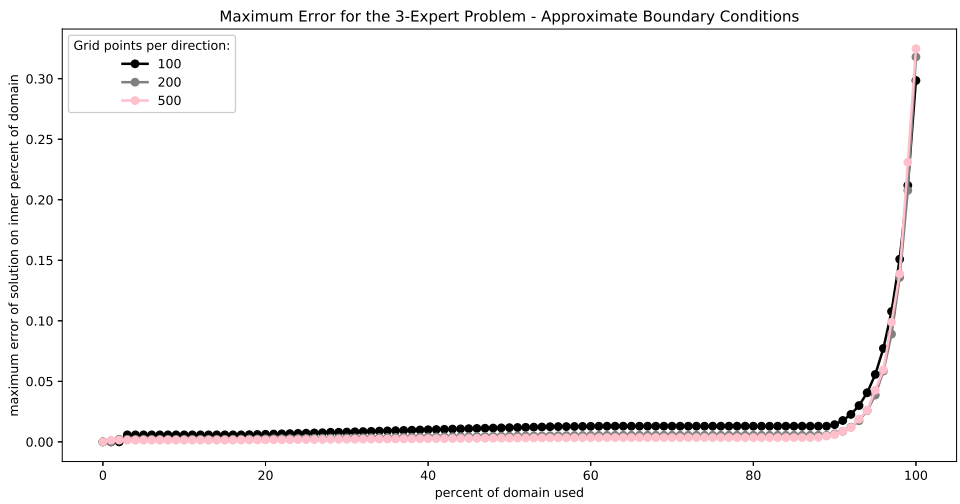
for the 3-expert problem. We observed that the numerical solution converged to the true solution for approximately the inner 90% of the domain, as did the derivatives. This provided justification to reliably use the inner portions of the domain in the following sections to assess optimal strategies.

5.1.2 Optimal Strategies

Optimal Strategy Data

In the plots in this section, we show the optimal strategies for $n = 3, 4, 5, 6, 7$ experts over different portions of the domain away from the approximate boundary condition where $x_1 = T$, where the approximate boundary condition $w = T$ is imposed.

Justification for evaluating the optimality at various interior segments of the domain can be found in Thm. 3.1.19 and Section 5.1.1. In particular, when we impose the approximate



(a) Error vs. percent of domain, 3 experts

Figure 5.2: In both of these plots, we can see that the boundary condition affects the solution, but only within a percentage of gridpoints close to the boundary.

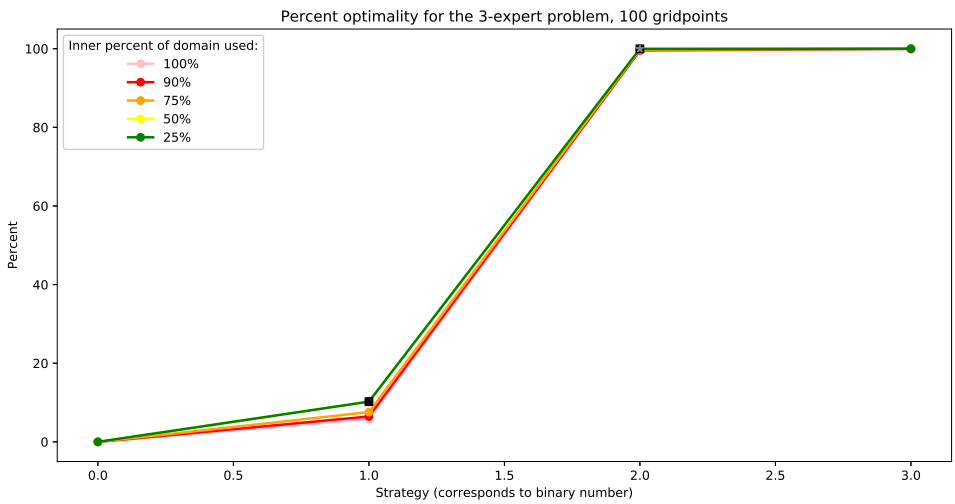
boundary condition at $x_1 = T$, the accuracy of w and its derivatives will be affected close to $x_1 = T$, but not within an inner percentage of the domain. The inner $K\%$ of the domain is defined in (5.1.3).

In order to measure which strategies might be optimal based on numerical data, we can compare the values of the numerical approximations of the directional derivatives. If a given directional derivative appears to be greater than others, we have reason to believe that the direction vector of that derivative may be an optimal strategy. To measure whether this may be the case for a given direction, we use the following statistic: Ratio of the sum over the domain of the derivative in the v_1 direction and the sum over the domain of the maximum directional derivative (for v_1 any fixed strategy in $0, 1^n$):

$$\text{opt}_{\mathbf{v}_1} = 100 \frac{\sum_{x \in \mathbb{D}_n^+} \mathbf{v}_1{}^T \nabla^2 u \mathbf{v}_1}{\sum_{x \in \mathbb{D}_n^+} \max_{\mathbf{v} \in \{0,1\}^n} \mathbf{v}^T \nabla^2 u \mathbf{v}} \tag{5.1.4}$$

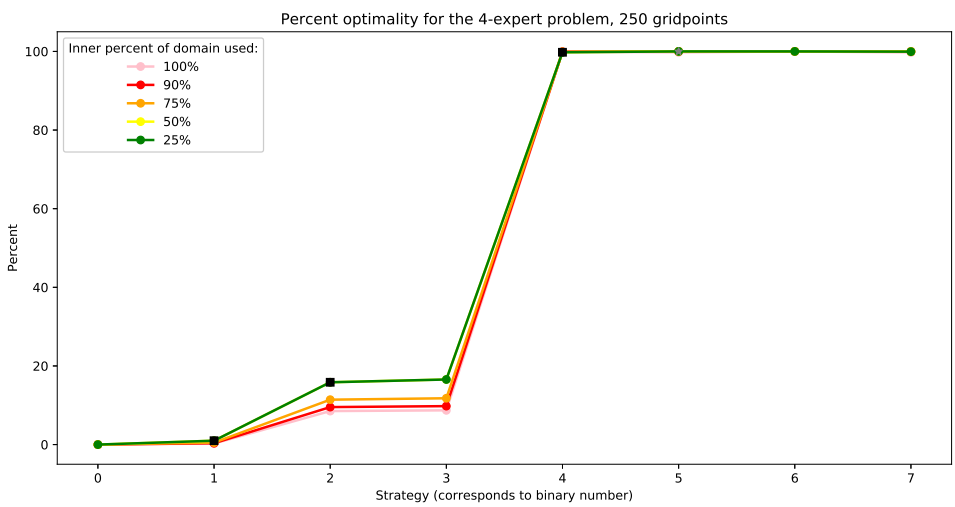
If a strategy $\mathbf{v}_1$ is optimal over the entire domain, then this ratio will be close to 100.

Based on the plots of $\text{opt}_{\mathbf{v}_1}$ vs. strategy $\mathbf{v}_1$ below, it may be hard to distinguish between supposed optimal and non-optimal strategies, since many have $\text{opt}_{\mathbf{v}_1}$ close to 100. Section 5.1.2 explains how we can make this distinction.



(a) Optimal strategies, 3 experts

Figure 5.3



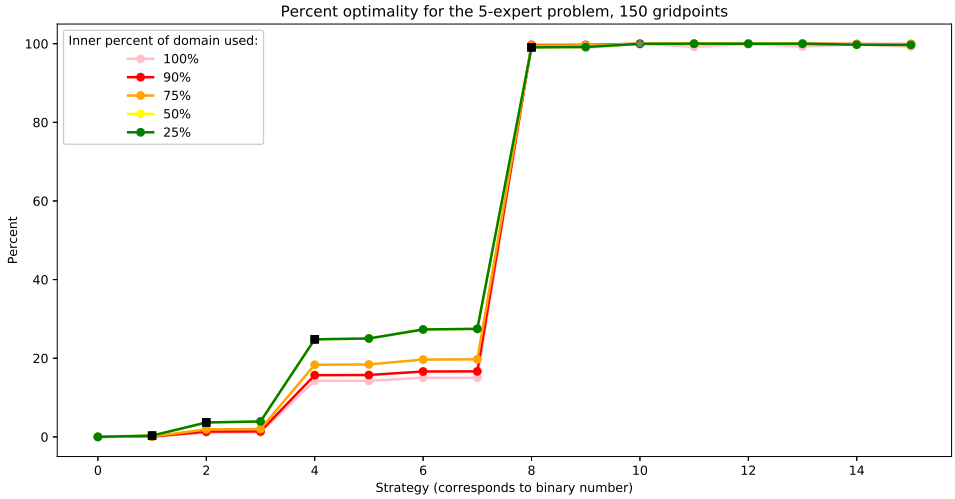
(a) Optimal strategies, 4 experts

Figure 5.4

Detecting Optimal Strategies

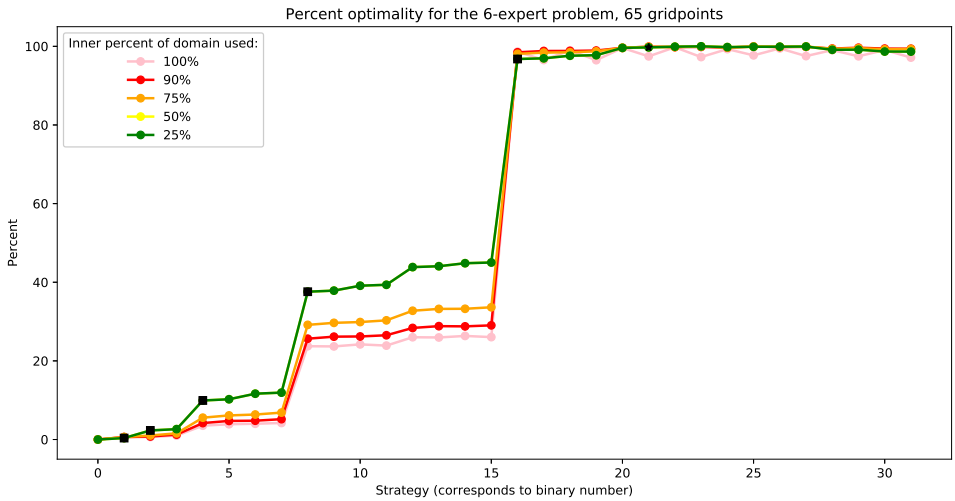
A major goal of the problem of prediction with expert advice is to find optimal strategies. Each strategy corresponds to one of the binary vectors v in the equation

$$u - \frac{1}{2}\max_{\mathbf{v}\in\{0,1\}^n} \mathbf{v}^T \nabla^2 u \mathbf{v} = g \qquad \text{on } \mathbb{R}^n \tag{5.1.5}$$



(a) Optimal strategies, 5 experts

Figure 5.5

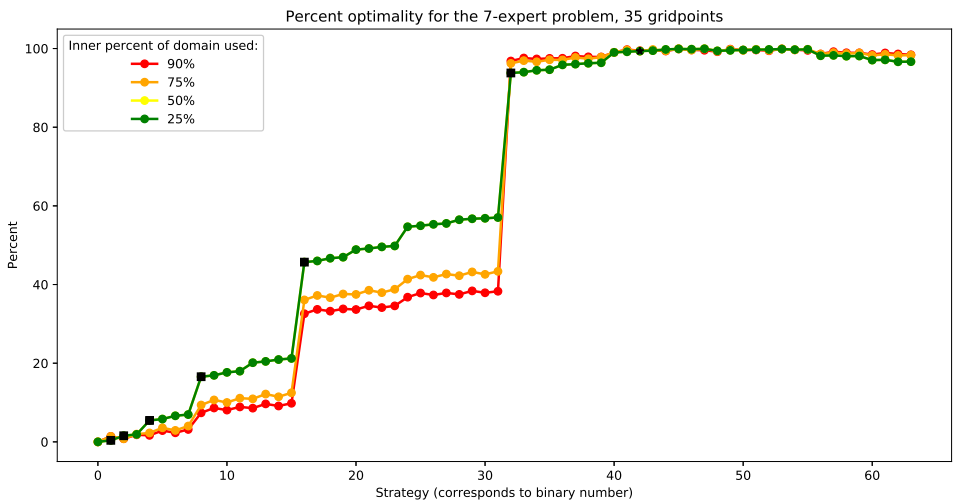


(a) Optimal strategies, 6 experts

Figure 5.6

for $g(x_1, ..., x_n) = \max\{x_1, ..., x_n\}$ and a strategy v_1 is considered optimal at a given point $x \in \mathbb{D}_n^+$ if ${v_1}^T D^2 u v_1 \geq \mathbf{v}^T \nabla^2 u \mathbf{v}$ at x, for all $v \in \{0,1\}^n$, $v_1 \neq v$.

In order to measure which strategies might be optimal based on numerical data, we can compare the values of the numerical approximations of the directional derivatives. If a given directional derivative appears to be greater than others, we have reason to believe that the



(a) Optimal strategies, 7 experts

Figure 5.7

Strategy	Slope of log-log plot
2	2.79703786
3	1.97474164

Table 5.1: Slopes for 3 experts. This supports the earlier finding that strategies 2 and 3 are optimal.

direction vector of that derivative may be an optimal strategy. To measure whether this may be the case for a given direction, we use the following statistic:

$$\text{opt}_{\mathbf{v}_1} = 100 \frac{\sum_{x \in \mathbb{D}_n^+} \mathbf{v}_1{}^T \nabla^2 u \mathbf{v}_1}{\sum_{x \in \mathbb{D}_n^+} \max_{\mathbf{v} \in \{0,1\}^n} \mathbf{v}^T \nabla^2 u \mathbf{v}} \tag{5.1.6}$$

If a strategy $\mathbf{v}_1$ is optimal over the entire domain, then this ratio will be close to 100. In particular, as we refine the grid resolution h, an optimal strategy $\mathbf{v}_1$ should obey the following criterion:

$$100 - \text{opt}_{v_1} \to 0 \text{ as } h \to 0. \tag{5.1.7}$$

The following tables show the slopes A received from doing a polynomial fit on a log-log plot of h vs. opt_{v_1}, assuming $\log(100 - \text{opt}_{v_1}) = A\log(h) + B$ for strategies v_1 that have appeared close to optimal (opt_{v_1} close to 100). A positive slope would indicate potential optimality, as it would suggest that (5.1.7) is being met.

Strategy	Slope of log-log plot
4	0.97744307
5	1.93527854
6	3.96529544
7	1.22441254

Table 5.2: Slopes for 4 experts. This supports the earlier finding that strategies 5 and 6 are optimal,and presents the possibility that strategies 4 and 7 may be optimal, or may be converging to slightly-suboptimal levels.

Strategy	Slope of log-log plot
8	1.758185
9	0.90888598
10	5.00197714
11	7.53416923
12	6.46732257
13	4.21853724
14	2.41076054
15	1.63424875

Table 5.3: Slopes for 5 experts. This indicates that several strategies from 8 to 15 may either be globally optimal, or converging to a level close to optimality.

Strategy	Slope of log-log plot
16	1.000367434653394
17	0.4698266224583626
18	0.8560419993676865
19	0.6565344550202243
20	2.1744920010693214
21	1.2255853055136015
22	2.4126302194454303
23	2.214054025816618
24	2.413565933585827
25	1.219850782331318
26	2.619852827816759
27	2.2012405893423113
28	1.6709691845424544
29	0.8677676177939316
30	1.2891070556445245
31	1.1675769514090613

Table 5.4: Slopes for 6 experts. This indicates potential optimality of strategies 20, 22-24, 26, 27.

We can use the slopes of the fitted log-log plots to get a sense of whether, for optimal-looking strategies, $100 - \text{opt}(\mathbf{v}_1)$ is converging to 0. For 3 and 4 experts, the results in this paper confirm the known optimal strategies. For 5, 6, and 7 experts, we observe several potentially-optimal strategies based on the polynomial fitting (slopes ≥ 2 indicate convergence rates of h^2 or better), and some which may not be converging to 0.

Strategy	Slope of log-log plot
32	-0.6156436494271919
33	-2.071163680174291
34	-0.9386752101995304
35	-1.620108529111811
36	-0.1288364793734311
37	-1.7189278056687674
38	-0.4563162822130352
39	-1.0868011624662257
40	2.221246869859825
41	-2.0835525524360596
42	2.2956136561490537
43	-0.11683984790448164
44	4.956696481982287
45	-0.29068497175340985
46	4.794579753039059
47	1.0359588393901393
48	2.442982365168055
49	-1.3711207130071636
50	2.776761217650851
51	1.046951720099005
52	3.571343034778775
53	0.058642619102782984
54	2.7425108513841026
55	1.236042324829215
56	0.5736834327123738
57	-0.9642458601456639
58	-0.29127231042515117
59	-0.3449944997060651
60	-0.3536449665914396
61	-1.498104633455475
62	-1.1567397362910683
63	-0.8949070045701722

Table 5.5: Slopes for 7 experts. This indicates potential optimality of ths trategies 40, 42, 44, 46, 48, 50, 52, 54.

Chapter 6

Conclusion

This paper presented analysis of a numerical scheme to solve an elliptic equation related to the problem of prediction with expert advice. By overcoming the curse of dimensionality, the equation could be solved numerically in higher dimensions. Observations related to the derivatives of the solution confirmed known results about optimal strategies for 3 and 4 experts, and indicated that there may be multiple optimal strategies for higher numbers of experts.

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Appendix A

Glossary and Acronyms

A.1 Definition of viscosity solution

For convenience, we recall the definitions of viscosity solutions for a general second order nonlinear partial differential equation

$$H(\nabla^2 u, \nabla u, u, x) = 0 \quad \text{in } \mathcal{O}, \tag{A.1.1}$$

where H is continuous and $\mathcal{O} \subset \mathbb{R}^n$. Let $\mathrm{USC}(\mathcal{O})$ (resp. $\mathrm{LSC}(\mathcal{O})$) denote the collection of functions that are upper (resp. lower) semicontinuous at all points in $\mathcal{O}$. We make the following definitions.

We first recall the test function definition of viscosity solution.

Definition A.1.1 (Viscosity solution)**.** We say that $u \in \mathrm{USC}(\mathcal{O})$ is a *viscosity subsolution* of (A.1.1) if for every $x \in \mathcal{O}$ and every $\varphi \in C^\infty(\mathbb{R}^n)$ such that $u - \varphi$ has a local maximum at x with respect to $\mathcal{O}$

$$H(\nabla^2 \varphi(x), \nabla \varphi(x), u(x), x) \leq 0.$$

We will often say that $u \in \mathrm{USC}(\mathcal{O})$ is a viscosity solution of $H \leq 0$ in $\mathcal{O}$ when u is a viscosity subsolution of (A.1.1).

Similarly, we say that $u \in \mathrm{LSC}(\mathcal{O})$ is a *viscosity supersolution* of (A.1.1) if for every $x \in \mathcal{O}$ and every $\varphi \in C^\infty(\mathbb{R}^n)$ such that $u - \varphi$ has a local minimum at x with respect to $\mathcal{O}$

$$H(\nabla^2 \varphi(x), \nabla \varphi(x), u(x), x) \geq 0.$$

We also say that $u \in \mathrm{LSC}(\mathcal{O})$ is a viscosity solution of $H \geq 0$ in $\mathcal{O}$ when u is a viscosity

supersolution of (A.1.1).

Finally, we say u is *viscosity solution* of (A.1.1) if u is both a viscosity subsolution and a viscosity supersolution.

For more details on the rich theory viscosity solutions, we refer the reader to the user's guide [38] and [39].